

Trigonometry (9 th std)

Choose the Appropriate Answer

1.If $\sin 30^\circ = x$ and $\cos 60^\circ = y$, then $x^2 + y^2$ is

- (a) $1/2$ (b) 0 (c) $\sin 90^\circ$ (d) $\cos 90^\circ$

Ans: (a) $1/2$

$$x^2 + y^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \left(\frac{1}{4}\right) + \left(\frac{1}{4}\right) = \left(\frac{2}{4}\right) = \frac{1}{2}$$

2.If $\tan \theta = \cot 37^\circ$, then the value of θ is

- (a) 37° (b) 53° (c) 90° (d) 1°

Ans: (b) 53°

$$\tan \theta = \cot 37^\circ = \cot (90^\circ - 53^\circ) = \tan 53^\circ$$

3.The value of $\tan 72^\circ \cdot \tan 18^\circ$ is

- (a) 0 (b) 1 (c) 18° (d) 72°

Ans: (b) 1

$$\tan 72^\circ \cdot \tan (90^\circ - 72^\circ) = \tan 72^\circ \cdot \cot 72^\circ = \tan 72^\circ \times 1 / \tan 72^\circ = 1$$

4.The value of $(2 \tan 30^\circ) / (1 - \tan^2 30^\circ)$ is equal to

- (a) $\cos 60^\circ$ (b) $\sin 60^\circ$ (c) $\tan 60^\circ$ (d) $\sin 30^\circ$

Ans: (c) $\tan 60^\circ$

$$\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ} = \frac{2 \times \frac{1}{\sqrt{3}}}{1 - \left(\frac{1}{\sqrt{3}}\right)^2} = \frac{2}{\sqrt{3}} \div 1 - \frac{1}{3} = \frac{2}{\sqrt{3}} \div \frac{2}{3} = \frac{2}{\sqrt{3}} \times \frac{3}{2} = \frac{3}{\sqrt{3}} = \sqrt{3} = \tan 60^\circ$$

5.If $2 \sin 2\theta = \sqrt{3}$ then the value of θ is

- (a) 90° (b) 30° (c) 45° (d) 60°

Ans: (b) 30°

$$2 \sin 2\theta = \sqrt{3} \Rightarrow \sin 2\theta = \sqrt{3} / 2 \Rightarrow \sin 2\theta = \sin 60^\circ \Rightarrow 2\theta = 60^\circ \Rightarrow \theta = 30^\circ$$

6.The value of $3 \sin 70^\circ \sec 20^\circ + 2 \sin 39^\circ \sec 51^\circ$ is

- (a) 2 (b) 3 (c) 5 (d) 6

Ans: (c) 5

$$\begin{aligned} & 3 \sin 70^\circ \sec 20^\circ + 2 \sin 39^\circ \sec 51^\circ \\ &= 3 \cdot \sin 70^\circ \cdot \sec (90^\circ - 70^\circ) + 2 \sin 39^\circ \cdot \sec (90^\circ - 39^\circ) \\ &= 3 \cdot \sin 70^\circ \cdot \operatorname{cosec} 70^\circ + 2 \sin 39^\circ \cdot \operatorname{cosec} 39^\circ \\ &= 3 \cdot \sin 70^\circ \times \frac{1}{\sin 70^\circ} + 2 \sin 39^\circ \times \frac{1}{\sin 39^\circ} = 3 + 2 = 5 \end{aligned}$$

7. The value of $(1 - \tan^2 45^\circ) / (1 + \tan^2 45^\circ)$ is
- (a) 2 (b) 1 (c) 0 (d) $1/2$

Ans: (c) 0

$$\frac{1 - \tan^2 45^\circ}{1 + \tan^2 45^\circ} = \frac{1 - 1}{1 + 1} = \frac{0}{2} = 0$$

8. The value of $\operatorname{cosec} (70^\circ + \theta) - \sec (20^\circ + \theta) + \tan (65^\circ + \theta) - \cot (25^\circ + \theta)$ is
- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (a) 0

$$\begin{aligned} & \operatorname{cosec} (70^\circ + \theta) - \sec (20^\circ + \theta) + \tan (65^\circ + \theta) - \cot (25^\circ + \theta) \\ &= \sec [90^\circ - (70^\circ + \theta)] - \sec (20^\circ + \theta) + \tan (65^\circ + \theta) - \tan [90^\circ - (25^\circ + \theta)] \\ &= \sec (20^\circ + \theta) - \sec (20^\circ + \theta) + \tan (65^\circ + \theta) - \tan (65^\circ + \theta) = 0 \end{aligned}$$

9. The value of $\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots \tan 89^\circ$ is
- (a) 0 (b) 1 (c) 2 (d) $\sqrt{3}/2$

Ans: (b) 1

$$\begin{aligned} & \tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots \tan 89^\circ \\ &= \tan (90^\circ - 89^\circ) \cdot \tan (90^\circ - 88^\circ) \cdot \tan (90^\circ - 87^\circ) \dots \tan 45^\circ \dots \tan (89^\circ) \\ &= \cot 89^\circ \cdot \cot 88^\circ \cdot \cot 87^\circ \dots \tan 45^\circ \dots \tan 87^\circ \cdot \tan 88^\circ \cdot \tan 89^\circ = 1 \end{aligned}$$

10. Given that $\sin \alpha = 1/2$ and $\cos \beta = 1/2$, then the value of $\alpha + \beta$ is
- (a) 0° (b) 90° (c) 30° (d) 60°

Ans: (b) 90°

$$\begin{aligned} \sin \alpha &= \frac{1}{2} \Rightarrow \sin 30^\circ = \frac{1}{2} \therefore \alpha = 30^\circ \\ \cos \beta &= \frac{1}{2} \Rightarrow \cos 60^\circ = \frac{1}{2} \therefore \beta = 60^\circ \\ \therefore \alpha + \beta &= 30^\circ + 60^\circ = 90^\circ \end{aligned}$$

11. The value of $\operatorname{cosec}^2 60^\circ - 1$ is equal to
- (a) $\cos^2 60^\circ$ (b) $\cot^2 60^\circ$ (c) $\sec^2 60^\circ$ (d) $\tan^2 60^\circ$

Ans: (b) $\cot^2 60^\circ$

12. The value of $\cos 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ$ is equal to
- (a) $\operatorname{cosec} 90^\circ$ (b) $\tan 90^\circ$ (c) $\sin 30^\circ + \cos 30^\circ$ (d) $\cos 90^\circ$

Ans: (d) $\cos 90^\circ$

13. The value of $\sin 57^\circ \cos 33^\circ$ is
- (a) $\cot 63^\circ$ (b) $\tan 27^\circ$ (c) 1 (d) 0

Ans: (c) 1

14. If $3 \operatorname{cosec} 36^\circ = \sec 54^\circ$ then the value of x is
- (a) 0 (b) 1 (c) 13 (d) 34

Ans: (c) 13

15. If $\cos A \cos 30^\circ = \sqrt{3}/4$, then the measures of A is

- (a) 90° (b) 60° (c) 45° (d) 30°

Ans: (b) 60°

16. In triangle ABC, if $AB = BC$ and $\angle B = 70^\circ$, $\angle A$ will be:

- (a) 70° (b) 110° (c) 55° (d) 130°

Ans: (c) 55°

$\begin{aligned}\angle A + \angle B + \angle C &= 180^\circ \Rightarrow 2\angle A + \angle B = 180^\circ \\ 2\angle A &= 180 - \angle B = 180 - 70 = 110^\circ \Rightarrow \angle A = 55^\circ\end{aligned}$
--

17. A triangle in which two sides are equal is called _____ triangle.

- (a) Scalene (b) Equilateral (c) Isosceles (d) None of the above

Ans: (c) Isosceles

18. The angles opposite to equal sides of a triangle are _____ angles.

- (a) Equal (b) Unequal (c) Supplementary (d) Complementary

Ans: (a) Equal

19. If ABC is an equilateral triangle, then each angle equals to:

- (a) 90° (b) 180° (c) 120° (d) 60°

Ans: (d) 60°

20. In a right triangle, the longest side is:

- (a) Perpendicular (b) Hypotenuse (c) Base (d) None of the above

Ans: (b) Hypotenuse

21. In $\triangle PQR$, $\angle R = \angle P$ and $QR = 4$ cm and $PR = 5$ cm. Then the length of PQ is

- (a) 2 cm (b) 2.5 cm (c) 4 cm (d) 5 cm

Ans: (c) 4 cm

22. In $\triangle ABC$, $BC = AB$ and $\angle B = 80^\circ$. Then $\angle A$ is equal to

- (a) 40° (b) 50° (c) 80° (d) 100°

Ans: (b) 50°

$x + 80^\circ + x = 180^\circ \Rightarrow 2x + 80^\circ = 180^\circ \Rightarrow 2x = 100^\circ \Rightarrow x = 50^\circ$
--

23. Two sides of a triangle are of lengths 5 cm and 1.5 cm. The length of the third side of the triangle cannot be

- (a) 3.4 cm (b) 3.6 cm (c) 3.8 cm (d) 4.1 cm

Ans: (a) 3.4 cm

24. In $\triangle ABC$, $AB = AC$ and $\angle B = 50^\circ$. Then $\angle C$ is equal to

- (a) 40° (b) 50° (c) 80° (d) 130°

Ans: (b) 50°

25. It is given that $\triangle ABC$, $\angle B = 40^\circ$ and $\angle A = 80^\circ$. Then which of the following is true?

- (a) $\angle C = 70^\circ$ (b) $\angle C = 60^\circ$ (c) $\angle C = 65^\circ$ (d) $\angle C = 40^\circ$

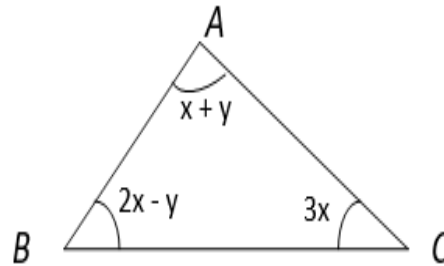
Ans: (b) $\angle C = 60^\circ$

26. A triangle has _____ vertices.

- (a) One (b) Two (c) Three (d) Four

Ans: (c) Three

27. Which of the following relation between x and y is correct?

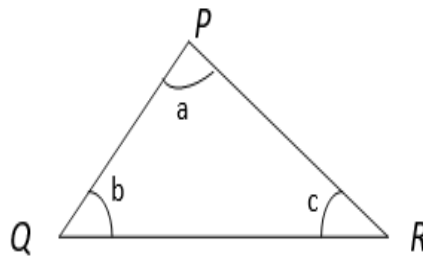


- (a) $x + y = 45^\circ$ (b) $x - y = 180^\circ$ (c) $3x + y = 90^\circ$ (d) $3x - y = 90^\circ$

Ans: (d) $3x - y = 90^\circ$

$$(x + y) + (2x - y) + (3x) = 180^\circ \Rightarrow 6x - 2y = 180^\circ \Rightarrow 3x - y = 90^\circ$$

28. Find the angles ($\angle P$, $\angle Q$, $\angle R$) of the triangle if $a + b = 120^\circ$ and $a - c = 30^\circ$.



- (a) $90^\circ, 30^\circ, 60^\circ$ (b) $90^\circ, 45^\circ, 45^\circ$ (c) $120^\circ, 30^\circ, 30^\circ$ (d) $90^\circ, 45^\circ, 80^\circ$

Ans: (a) $90^\circ, 30^\circ, 60^\circ$

29. Find the type of the $\triangle ABC$ if angle B and C are equal.

- (a) Right angled triangle (b) Equilateral triangle
(c) Obtuse angled triangle (d) Scalene triangle

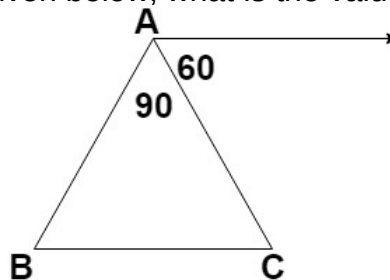
Ans: (b) Equilateral triangle

30. The sum of the angles of a triangle is equal to _____

- (a) 270° (b) 90° (c) 180° (d) 360°

Ans: (c) 180°

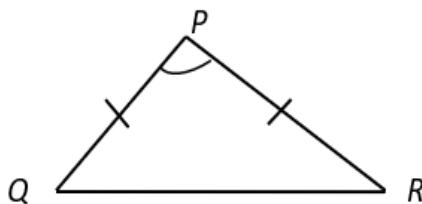
31. From the diagram given below, what is the value of $\angle CBA$?



- (a) 30° (b) 60° (c) 90° (d) 100°

Ans: (a) 30°

32. Find the angle $\angle Q$ and $\angle R$ if $\angle P = 80^\circ$ and $PQ = PR$.



- (a) 60° (b) 50° (c) 30° (d) 90°

Ans: (b) 50°

33. In a right-angled triangle, if one leg is 6 cm and the length of the hypotenuse is 10 cm, find the value of other leg?

- (a) 6 m (b) 7 m (c) 8 m (d) 4 m

Ans: (c) 8 m

$$b = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8 \text{ m}$$

34. Two poles of height 9m and 14m stand on a plane ground. If the distance between their feet is 12m, find the distance between their tips.

- (a) 12 cm (b) 13 cm (c) 15 cm (d) 11 cm

Ans: (b) 13 cm

$$\begin{aligned} \text{The horizontal distance is 12m, and the vertical difference is } 14 - 9 = 5\text{m.} \\ \text{Distance}^2 = 12^2 + 5^2 = 144 + 25 = 169. \text{ Distance} = \sqrt{169} = 13 \text{ m.} \end{aligned}$$

35. Which of triangle whose sides are given below are right angled?

- (a) $AB = 8 \text{ cm}$, $AC = 7 \text{ cm}$, $BC = 6 \text{ cm}$
 (b) $AB = 5 \text{ cm}$, $AC = 6 \text{ cm}$, $BC = 7 \text{ cm}$
 (c) $AC = 3 \text{ cm}$, $AB = 4 \text{ cm}$, $BC = 5 \text{ cm}$
 (d) $AC = 2 \text{ cm}$, $AB = 3 \text{ cm}$, $BC = 4 \text{ cm}$

Ans: (c) $AC = 3 \text{ cm}$, $AB = 4 \text{ cm}$, $BC = 5 \text{ cm}$

36. In a right-angled triangle, if one leg is 3 cm and the length of the hypotenuse is 5 cm, find the value of other leg?

- (a) 6 cm (b) 7 cm (c) 4 cm (d) 3 cm

Ans: (c) 4 cm

37. A ladder 10 m long is placed against a wall. The foot of the ladder is 8 m away from the wall. How high does the ladder reach on the wall?

- (a) 5 m (b) 6 m (c) 7 m (d) 9 m

Ans: (b) 6 m

$$8^2 + b^2 = 10^2 \Rightarrow 64 + b^2 = 100 \Rightarrow b^2 = 100 - 64 = 36 \Rightarrow b = \sqrt{36} = 6$$

38. In a right triangle ABC with right angle at B, if $AB = 12 \text{ cm}$ and $BC = 16 \text{ cm}$, find AC.

- (a) 15 cm (b) 18 cm (c) 20 cm (d) 17 cm

Ans: (c) 20 cm

$$AC^2 = (12)^2 + (16)^2 = 144 + 256 = 400 \Rightarrow AC = \sqrt{400} = 20$$

39. Find the hypotenuse of a triangle with sides 24 cm and 7 cm.

- (a) 23 cm (b) 20 cm (c) 22 cm (d) 25 cm

Ans: (d) 25 cm

$$c^2 = a^2 + b^2 \Rightarrow 24^2 + 7^2 = 576 + 49 = 625 \Rightarrow c = \sqrt{625} = 25$$

40. A 25 m ladder reaches a window 20 m above the ground. How far is the foot of the ladder from the wall?

- (a) 10 m (b) 12 m (c) 15 m (d) 13 m

Ans: (c) 15 m

$$25^2 = 20^2 + \text{base}^2 \Rightarrow 625 = 400 + \text{base}^2 \Rightarrow \text{base}^2 = 225. \text{Distance} = \sqrt{225} = 15 \text{ m.}$$

41. Check if sides 1.4 cm, 4.8 cm, and 5 cm form a right triangle?

- (a) True (b) False (c) Can't determine (d) None of these

Ans: (a) True

$$1.4^2 + 4.8^2 = 1.96 + 23.04 = 25. \text{ Since } 5^2 = 25, \text{ this is a right triangle.}$$

42. Find another side of a right-angled triangle whose hypotenuse is 13 cm and one side is 5 cm.

- (a) 14 cm (b) 15 cm (c) 12 cm (d) 9 cm

Ans: (c) 12 cm

$$\text{Other side} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \text{ cm.}$$

43. What is the value of $\tan 60^\circ / \cot 30^\circ$

- (a) 3 (b) 1 (c) 2 (d) 0

Ans. (b) 1

$$\tan 60^\circ / \cot 30^\circ = \sqrt{3} / \sqrt{3} = 1$$

44. $(\sin 30^\circ + \cos 60^\circ) - (\sin 60^\circ + \cos 30^\circ)$ equals

- (a) $1 + \sqrt{3}$ (b) 0 (c) $1 - \sqrt{3}$ (d) $1 + 2\sqrt{3}$

Ans: (c) $1 - \sqrt{3}$

$$(\frac{1}{2} + \frac{1}{2}) - (\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}) = 1 - (2 * \frac{\sqrt{3}}{2}) = 1 - \sqrt{3}$$

45. If $\cos X = 2/3$. Find the value of $\tan X$

- (a) $\sqrt{5}/2$ (b) $5/2$ (c) $5/\sqrt{2}$ (d) $\sqrt{5/2}$

Ans (a) $\sqrt{5}/2$

$$\begin{aligned} 1 + \tan^2 x &= \sec^2 x \Rightarrow \cos x = 1 / \sec x \Rightarrow 1 + \tan^2 x = 1 / \cos^2 x \\ \tan^2 x &= (1 / \cos^2 x) - 1 \Rightarrow \tan x = \sqrt{((1 / \cos^2 x) - 1)} \Rightarrow \tan x = \sqrt{(9/4) - 1} \\ \tan x &= \sqrt{(5/4)} \Rightarrow \tan x = \sqrt{5}/2 \end{aligned}$$

46. If $\cos x = a / b$, find the value of $\sin x$
 (a) $\sqrt{(b - a)} / a$ (b) $(b^2 - a^2) / b$ (c) $\sqrt{(b^2 - a^2)} / b$ (d) $(b - a) / a$
Ans: (c) $\sqrt{(b^2 - a^2)} / b$

47. $\tan 30^\circ / (1 + \tan^2 30^\circ)$ is equal to
 (a) $\sin 30^\circ$ (b) $\sin 60^\circ$ (c) $\cos 30^\circ$ (d) $\cos 60^\circ$
Ans (b) $\sin 60^\circ$

48. Find the value of $(\sin 45^\circ + \cos 45^\circ)$
 (a) $1/\sqrt{2}$ (b) $1/2$ (c) $\sqrt{3}/2$ (d) $\sqrt{2}$
Ans: (d) $\sqrt{2}$

49. If the value of $\sin A = 1/2$. Find the value of $\cot A$.
 (a) $1 + \sqrt{3}$ (b) 0 (c) $1 - \sqrt{3}$ (d) $1 + 2\sqrt{3}$
Ans. (c) $1 - \sqrt{3}$

$$\left(\frac{1}{2} + \frac{1}{2} \right) - \left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right) = 1 - (2 * \sqrt{3} / 2) = 1 - \sqrt{3}$$

50. If a triangle ABC is right-angled at C. What will be the value of $\cos (A+B)$
 (a) 1 (b) 0 (c) $\sqrt{3}/2$ (d) $1/2$
Ans: (b) 0

51. Find the value of $(\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ)$ is
 (a) $1/2$ (b) 1 (c) 0 (d) 2
Ans: (b) 1

52. Find the value of expression $[\operatorname{cosec} (75^\circ + \theta) - \sec (15^\circ - \theta) - \tan (55^\circ + \theta) + \cot (35^\circ - \theta)]$
 (a) 0 (b) 1 (c) -1 (d) $3/2$
Ans: (a) 0

53. If $\cos (\alpha + \beta) = 0$, then $\sin (\alpha - \beta)$ can be reduced to
 (a) $\cos \beta$ (b) $\cos 2\beta$ (c) $\sin \alpha$ (d) $\sin 2\alpha$
Ans: (b) $\cos 2\beta$

54. If $\sin A + \sin 2A = 1$, then the value of expression $(\cos 2A + \cos 4A)$
 (a) 1 (b) 2 (c) 3 (d) $1/2$
Ans: (a) 1

55. If $\cos 9\alpha = \sin \alpha$ and $9\alpha < 90^\circ$. Find the value of $\tan 5\alpha$
 (a) 0 (b) $\sqrt{3}$ (c) $1/\sqrt{3}$ (d) 1
Ans: (d) 1

56. Find the value of the expression $(\sin 6\theta + \cos 6\theta + 3 \sin 2\theta \cos 2\theta)$
 (a) 0 (b) 1 (c) 2 (d) 3
Ans: (b) 1

57. $\sin 2A = 2\sin A$ only when the value of A is
 (a) 0° (b) 45° (c) 60° (d) 30°
Ans: (a) 0°

58. In a $\triangle ABC$, right angled at B, $AB = 24$ cm, $BC = 7$ cm. Find the value of $\tan C$
(a) $12/7$ (b) $24/7$ (c) $7/24$ (d) $20/7$

Ans: (b) $24/7$

59. If $\sin A = 3/5$, where A is an acute angle, the $\cos A$ is:

(a) $4/5$ (b) $3/4$ (c) $5/3$ (d) $5/4$

Ans: (a) $4/5$

60. The value of $\tan 45^\circ$ is:

(a) 0 (b) $1/\sqrt{3}$ (c) 1 (d) $\sqrt{3}$

Ans: (c) 1

61. If $\sin \theta = 0$, then θ equals:

(a) 30° (b) 45° (c) 60° (d) 0°

Ans: (d) 0

62. The reciprocal of $\cos A$ is:

(a) $\sin A$ (b) $\tan A$ (c) $\sec A$ (d) $\operatorname{cosec} A$

Ans: (c) $\sec A$

63. The value of $\sin 30^\circ$ is:

(a) 1 (b) $\sqrt{3}/2$ (c) $1/2$ (d) 0

Ans: (c) $1/2$

64. If $\tan A = 1$, where A is acute, then A is:

(a) 30° (b) 45° (c) 60° (d) 90°

Ans: (b) 45°

65. In a right-angled triangle, the ratio of the opposite side to the hypotenuse is called:

(a) Cosine (b) Tangent (c) Sine (d) Secant

Ans: (c) Sine

66. If $\cos A = 5/13$, then $\sin A$ is

(a) $12/13$ (b) $5/12$ (c) $13/5$ (d) $12/5$

Ans: (a) $12/13$

67. The value of $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$ is:

(a) 2 (b) 0 (c) 1 (d) -1

Ans: (c) 1

68. If $\sin A + \operatorname{cosec} A = 2$, then $\sin^2 A + \operatorname{cosec}^2 A$ is:

(a) 1 (b) 2 (c) 3 (d) 4

Ans: (b) 2

69. If $4 \cos^2 x = 3$ and x is an acute angle, the value of x is:

(a) 30° (b) 45° (c) 60° (d) 90°

Ans: (a) 30°

70.If $\sin A = 1/2$, then the value of $\cos A$ is:

- (a) $1/\sqrt{2}$ (b) $\sqrt{3}/2$ (c) $\sqrt{3}/1$ (d) 1

Ans: (b) $\sqrt{3}/2$

71.The value of $\tan 45^\circ + \cot 45^\circ$ is:

- (a) 1 (b) 0 (c) 2 (d) $\sqrt{2}$

Ans: (c) 2

72.In a right angled triangle ABC, right - angled at B, if $AB = 3$ cm and $AC = 6$ cm, then $\angle ACB$ is:

- (a) 30° (b) 45° (c) 60° (d) 90°

Ans: (a) 30°

73.The value of $(\sin 30^\circ + \cos 30^\circ) - (\sin 60^\circ + \cos 60^\circ)$ is:

- (a) -1 (b) 0 (c) 1 (d) 2

Ans: (b) 0

74.If $\sec A + \tan A = y$, then $\sec A - \tan A$ is:

- (a) y (b) 2y (c) $1/y$ (d) 1

Ans: (c) $1/y$

75.In a right angled triangle, the sine of an angle is defined as:

- (a) $\frac{\text{Adjacent}}{\text{Hypotenuse}}$ (b) $\frac{\text{Opposite}}{\text{Adjacent}}$ (c) $\frac{\text{Opposite}}{\text{Hypotenuse}}$ (d) $\frac{\text{Hypotenuse}}{\text{Opposite}}$

Ans: (c) $\frac{\text{Opposite}}{\text{Hypotenuse}}$

76.Which of the following is equivalent to $\tan \theta =$

- (a) $\frac{\cos \theta}{\sin \theta}$ (b) $\frac{\sin \theta}{\cos \theta}$ (c) $\frac{1}{\cos \theta}$ (d) $\frac{1}{\sin \theta}$

Ans: (b) $\frac{\sin \theta}{\cos \theta}$

77.If $\cos \theta = 4/5$, then $\sin \theta$ is:

- (a) $3/5$ (b) $5/4$ (c) $3/4$ (d) $1/5$

Ans: (a) $3/5$

$$\sin^2 \theta = 1 - \cos^2 \theta = 1 - \frac{16}{25} = \frac{9}{25}$$

78.The value of $\sin 30^\circ$ is:

- (a) $\frac{\sqrt{3}}{2}$ (b) 1 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{2}$

Ans: (d) $\frac{1}{2}$

79.If $\sin A = 1 / \sqrt{2}$, then the magnitude of angle A is:

- (a) 30° (b) 60° (c) 90° (d) 45°

Ans: (d) 45°

80.The value of $(\cot^2 45^\circ - 1)$ is:

- (a) 1 (b) 0 (c) -1 (d) 2

Ans: (b) 0

81.In triangle PQR, right – angled at Q, PQ = 3 cm and PR = 6 cm, Find the $\angle P$.

- (a) 30° (b) 45° (c) 60° (d) 90°

Ans: (c) 60°

82.Which of the following is true?

- (a) $\sin(90^\circ - \theta) = \sin \theta$ (b) $\sin(90^\circ - \theta) = \cos \theta$
(c) $\cos(90^\circ - \theta) = \cos \theta$ (d) $\tan(90^\circ - \theta) = \tan \theta$

Ans: (b) $\sin(90^\circ - \theta) = \cos \theta$

83.The value of $\sin 60^\circ / \cos 30^\circ$ is:

- (a) $1 / 2$ (b) 1 (c) $\sqrt{3} / 2$ (d) $1 / \sqrt{3}$

Ans: (b) 1

$$\boxed{\frac{\sin 60^\circ}{\cos 30^\circ} = \frac{\sqrt{3}/2}{\sqrt{3}/2} = 1}$$

84.The reciprocal of $\cos A$ is:

- (a) $\sin A$ (b) $\tan A$ (c) $\sec A$ (d) $\operatorname{cosec} A$

Ans: (c) $\sec A$

85.if $4 \cos^2 x = 3$ and x is an acute angle, the value of x is:

- (a) 30° (b) 45° (c) 60° (d) 90°

Ans: (a) 30°

86.The value of $(\sec A + \tan A)(1 - \sin A)$ is equal to:

- (a) $\sec A$ (b) $\sin A$ (c) $\operatorname{cosec} A$ (d) $\cos A$

Ans: (d) $\cos A$

87.In a right-angled triangle, the ratio of the opposite side to the hypotenuse is called:

- (a) Cosine (b) Tangent (c) Sine (d) Secant

Ans: (c) Sine

88.If $\sin A + \operatorname{cosec} A = 2$, find the value of $\sin^2 A + \operatorname{cosec}^2 A$.

- (a) 2 (b) 4 (c) 1 (d) 0

Ans: (a) 2

89. Which of the following is the correct definition of tangent in a right triangle?

- (a) $\frac{\text{Adjacent}}{\text{Hypotenuse}}$ (b) $\frac{\text{Opposite}}{\text{Hypotenuse}}$ (c) $\frac{\text{Opposite}}{\text{Adjacent}}$ (d) $\frac{\text{Adjacent}}{\text{Opposite}}$

Ans: C) $\frac{\text{Opposite}}{\text{Adjacent}}$

90. $2 \tan 45^\circ + \cos 45^\circ - \sin 45^\circ = ?$

- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (c) 2

91. $2 \tan 45^\circ + \cos 30^\circ - \sin 60^\circ =$

- (a) 2 (b) 3 (c) 10 (d) -2

Ans: (a) 2

92. $\cos 60^\circ * \cos 30^\circ + \sin 60^\circ \sin 30^\circ$

- (a) $1/2$ (b) $1/\sqrt{2}$ (c) $\sqrt{3}/2$ (d) $\sqrt{3}$

Ans: (c) $\sqrt{3}/2$

93. $\tan 60^\circ / (\sin 60^\circ + \cos 60^\circ)$

- (a) $\sqrt{3}/2$ (b) $3 - \sqrt{3}$ (c) $\sqrt{3}$ (d) $1/\sqrt{3}$

Ans: (b) $3 - \sqrt{3}$

94. If $\tan = 7/8$, then the value of $[(1 + \sin)(1 - \sin)] / [(1 + \cos)(1 - \cos)]$.

- (a) $\frac{7}{8}$ (b) $\frac{8}{7}$ (c) $\frac{49}{64}$ (d) $\frac{64}{49}$

Ans: D) $\frac{64}{49}$

95. If $2\sin 2\theta = \sqrt{3}$, then the value of θ is

- (a) 0° (b) 30° (c) 45° (d) 60°

Ans: (b) 30°

96. Given that $\sin \alpha = 1/2$ and $\cos \beta = 1/2$, then the value of $\alpha + \beta$ is

- (a) 30° (b) 60° (c) 90° (d) 45°

Ans: (c) 90°

97. The value of $2\tan 30^\circ / 1 - \tan^2 30^\circ$ is equal to

- (a) $1/2$ (b) $\sqrt{3}/2$ (c) $\sqrt{3}$ (d) $1/\sqrt{2}$

Ans: (c) $\sqrt{3}$

98. If $\sin 30^\circ = x$ and $\cos 60^\circ = y$, then $x^2 + y^2$ is

- (a) $1/2$ (b) $\cos 90^\circ$ (c) 0 (d) $\sin 90^\circ$

Ans: (a) $1/2$

$x = \sin 30^\circ = \frac{1}{2} \text{ and } y = \cos 60^\circ = \frac{1}{2}$ $x^2 + y^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
--

99.If $3\cos 2\theta = \sqrt{3} / 2$, then the value of θ is

- (a) 20° (b) 15° (c) 10° (d) 25°

Ans: (b) 15°

$$\cos 2\theta = \frac{\sqrt{3}}{2} \Rightarrow 2\theta = 30^\circ \Rightarrow \theta = 15^\circ$$

100.The lengths of the sides of some triangles in some unit are given below.

Which of them is a right-angled triangle?

- (a) 7, 9, 13 (b) 10, 24, 26 (c) 6, 8, 12 (d) 8, 12, 16

Ans: (b) 10, 24, 26

$$c^2 = a^2 + b^2 \Rightarrow 26^2 = 10^2 + 24^2 \Rightarrow 676 = 100 + 576 \Rightarrow 676 = 676$$